



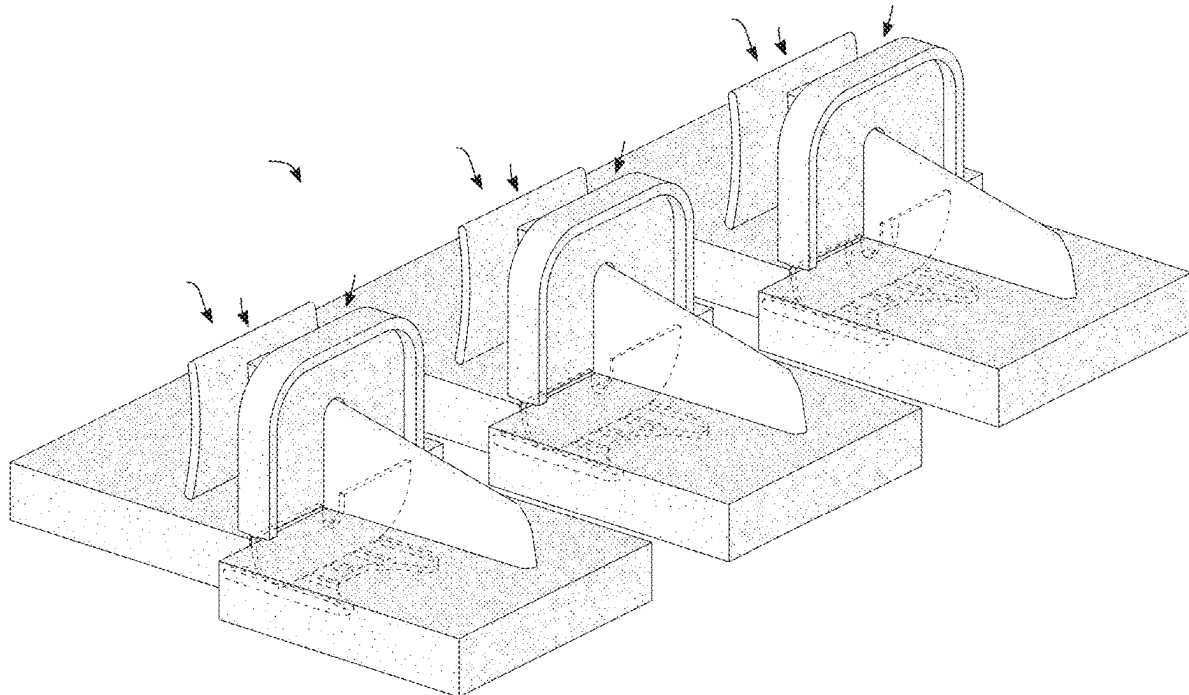
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(19) **United States**(12) **Patent Application Publication****Toca et al.**(10) **Pub. No.: US 2025/0277378 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **TILE-SETTING LIPPAGE AND SPACING CONTROL**(52) **U.S. Cl.**
CPC **E04F 21/0092** (2013.01); **E04F 21/22** (2013.01)(71) Applicant: **DTM INNOVATIONS LLC**, Metairie, LA (US)(72) Inventors: **Roland Toca**, New Orleans, LA (US);
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Darrin Godbold, Metairie, LA (US)(73) Assignee: **DTM INNOVATIONS LLC**, Metairie, LA (US)(21) Appl. No.: **19/197,769**(22) Filed: **May 2, 2025****Related U.S. Application Data**

(63) Continuation-in-part of application No. 18/534,039, filed on Dec. 8, 2023, which is a continuation-in-part of application No. 17/824,761, filed on May 25, 2022, now Pat. No. 11,840,852.

Publication Classification(51) **Int. Cl.**
E04F 21/00 (2006.01)
E04F 21/22 (2006.01)(57) **ABSTRACT**

A tile-setting lippage and spacing control system and method for use with tiles set with mortar as floors or walls on prepared surfaces, providing spacing tabs each having a tab body with a tab opening to accommodate a wedge, a strengthened tab border, and a lower tab body meant to extend downward between tiles as a spacer. A foot unit having a footplate and footplate ridges is removably attached at the bottom of the lower tab body through tab-to-foot joints at the footplate ridges. In use, mortar is applied to a prepared surface. Tile is placed with spacing tabs placed into the grout-line gaps between tiles, with the footplate of the foot unit extending underneath both tiles and embedded in the mortar. The lower tab body extends upward between the tiles and sets the gap or grout line between the tiles. The tab body and strengthened tab border extend upward above the tops of the tiles. A wedge is pushed into the tab opening, placing downward pressure on the tops of the tiles while pulling upward against the foot unit locked underneath the tiles, creating a clamping force which holds the tops of the tiles level with each other. The tab-to-foot joints withstand the straight-line constant clamping force between the wedge and the foot unit during use, and then break when a sharp oblique force is applied, allowing removal from the gap between tiles without damage to the tiles. After the mortar cures and sets the spacing tab is removed by applying a sharp oblique force, tearing the tab body cleanly away from the foot unit which remains embedded in mortar underneath the tiles.



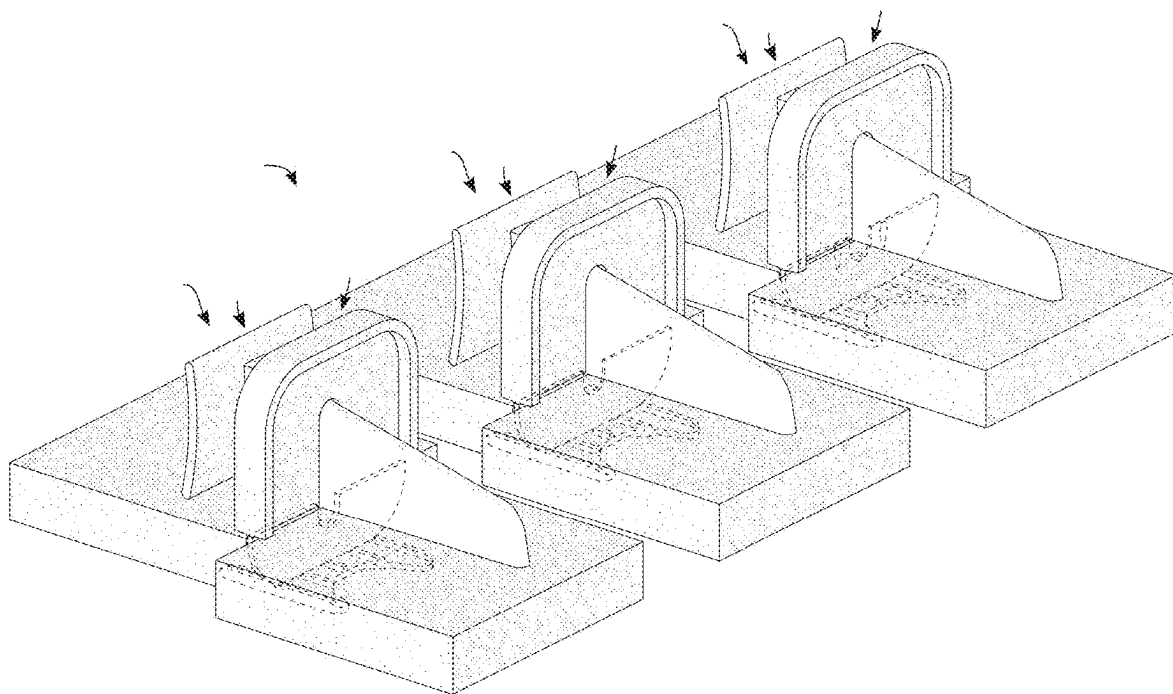


FIG. 1

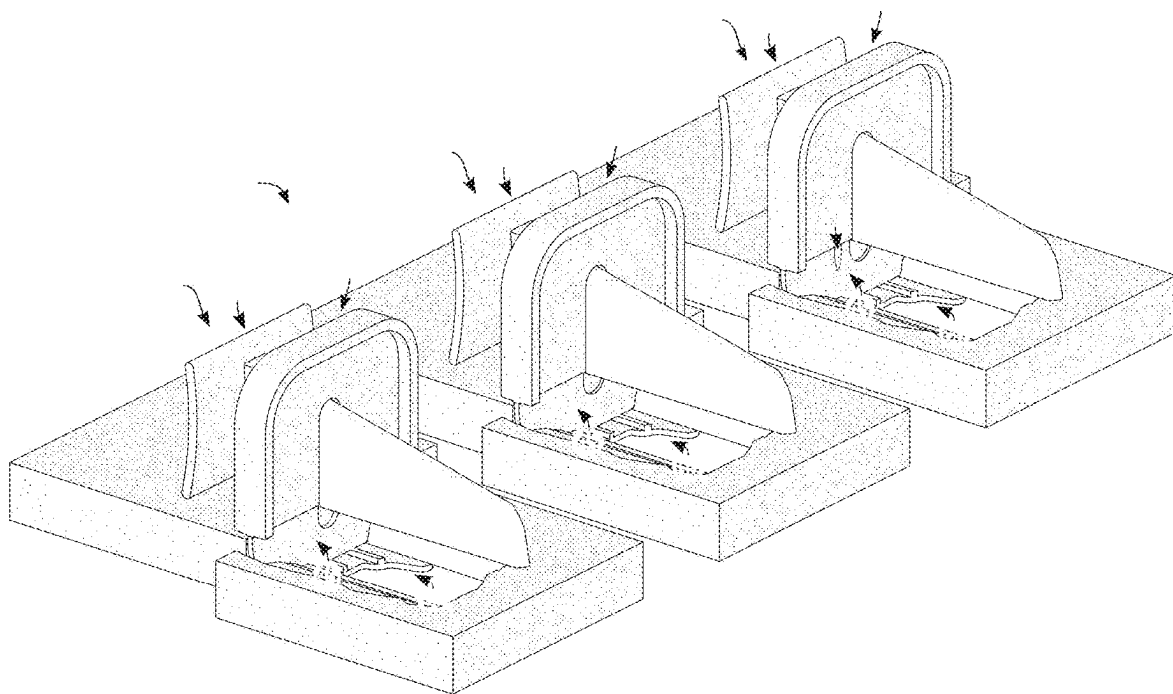


FIG. 2

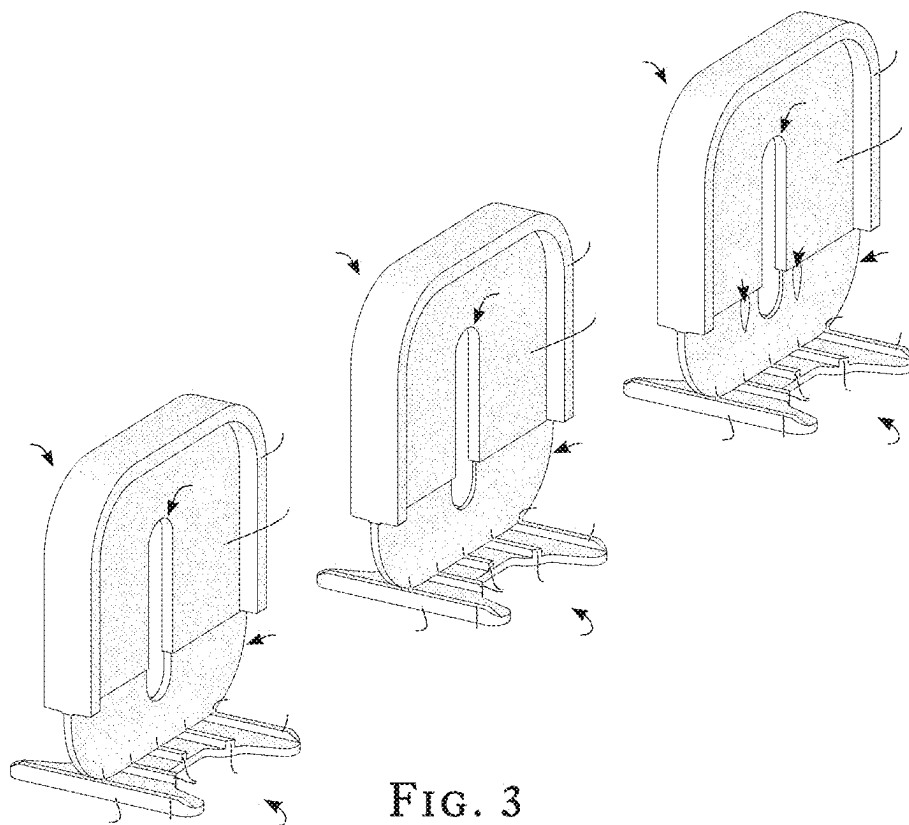


FIG. 3

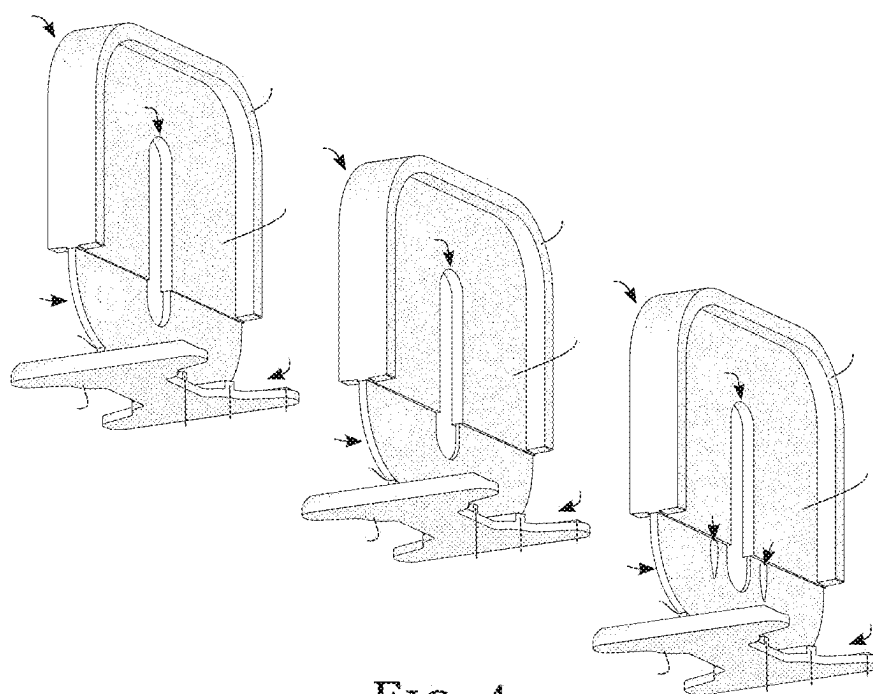


FIG. 4

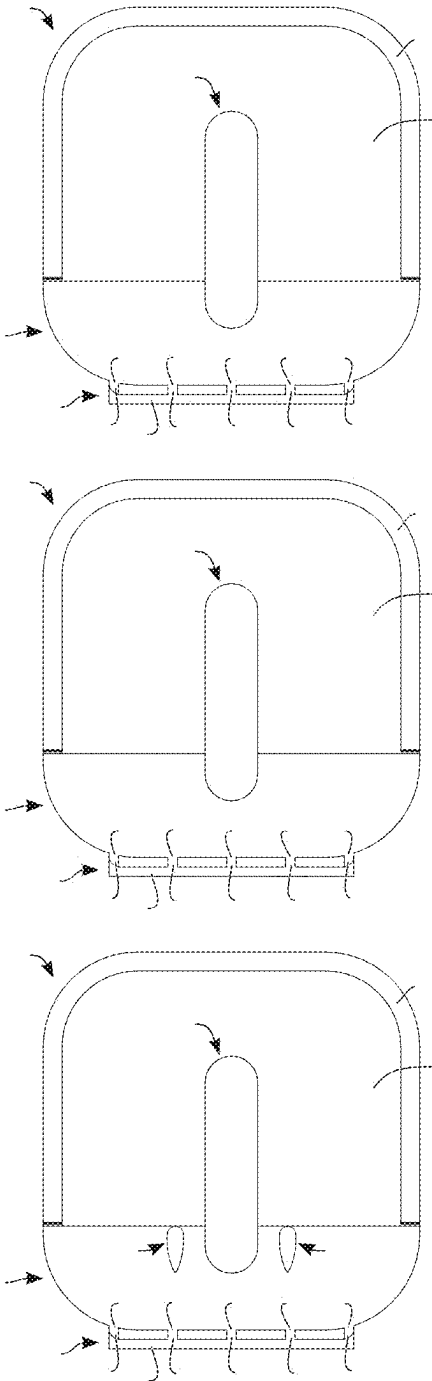


FIG. 5

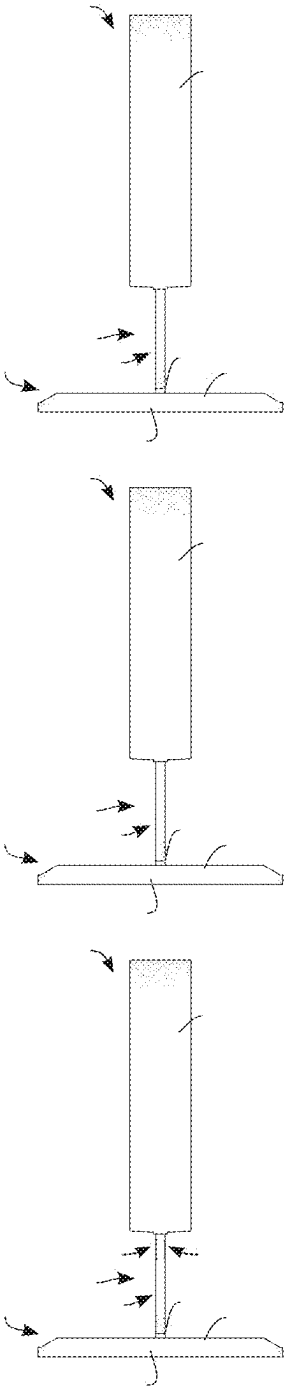


FIG. 6

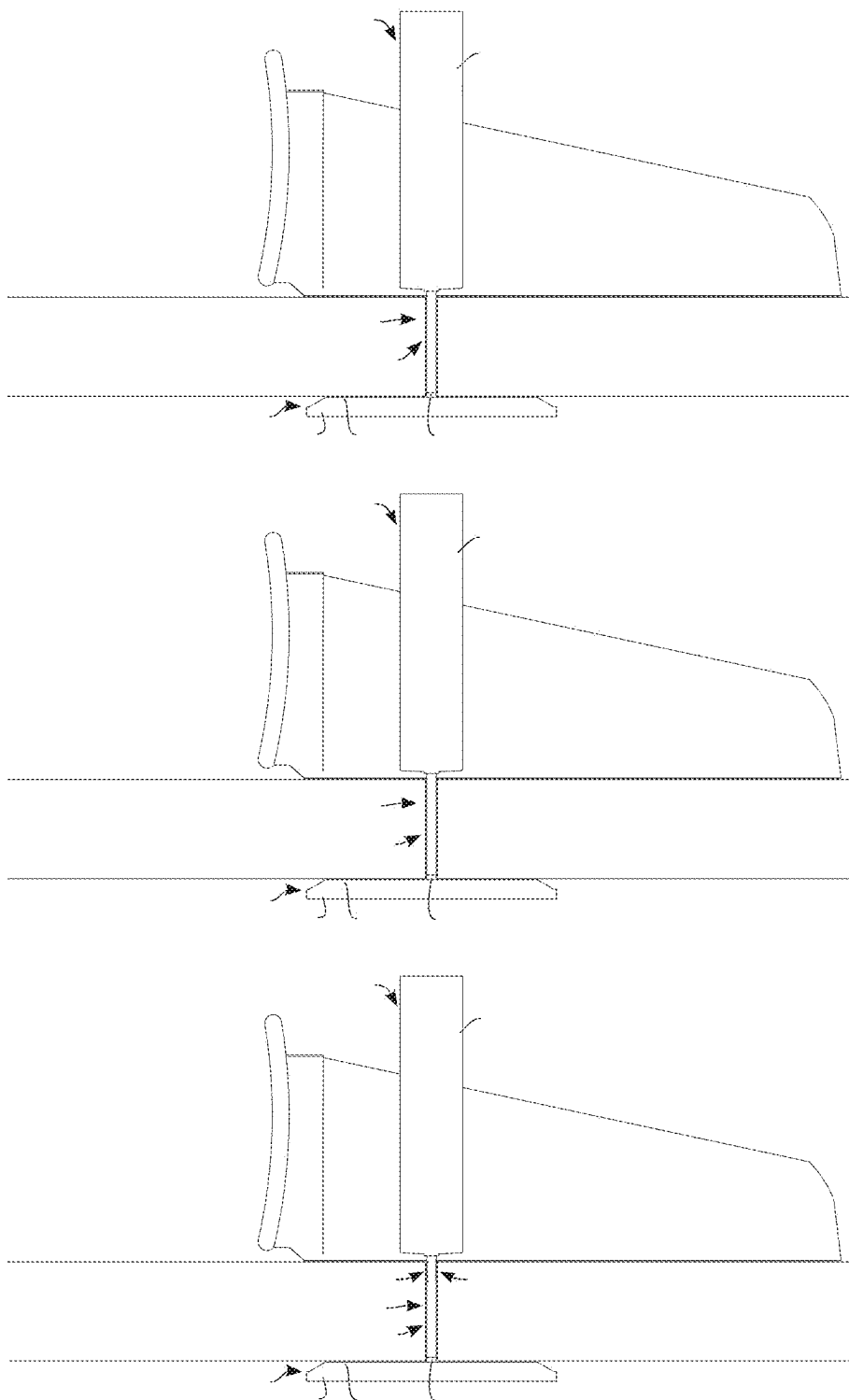


FIG. 7

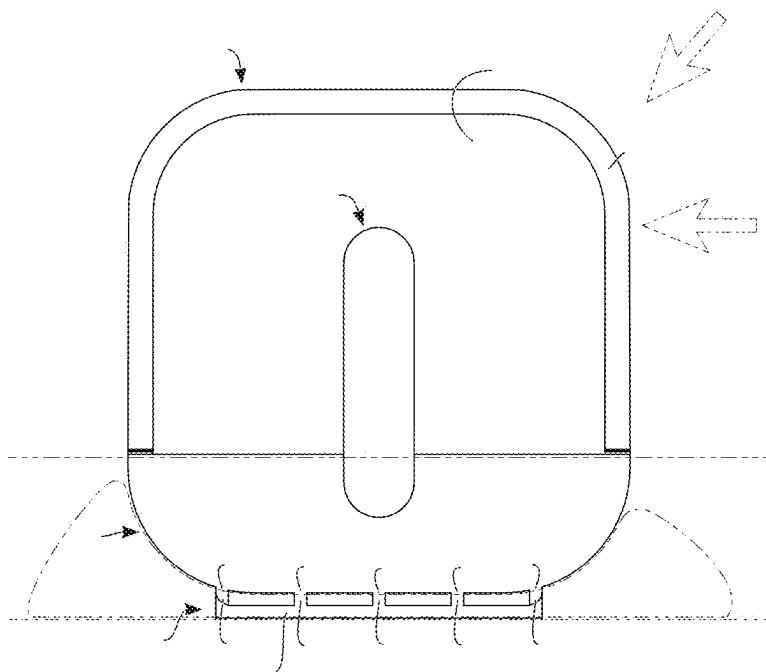


FIG. 8

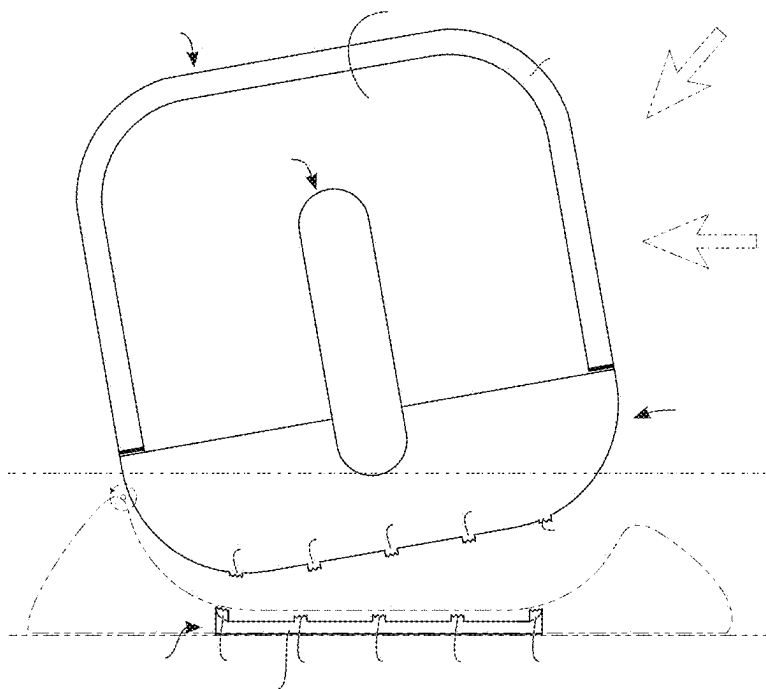


FIG. 9

TILE-SETTING LIPPAGE AND SPACING CONTROL

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation-in-part of co-pending application Ser. No. 18/534,039, filed on Dec. 8, 2023, which is a continuation-in-part of application Ser. No. 17/824,761, filed on May 25, 2022, now U.S. Pat. No. 11,840,852, issued on Dec. 12, 2023, the full disclosures of which are incorporated by reference herein and priority of which are hereby claimed.

BACKGROUND OF THE INVENTION

[0002] This invention provides a tile-setting lippage and spacing control system and method for use with tiles set with mortar as floors or walls on prepared surfaces.

[0003] When setting tiles such as gauged porcelain tiles, panels, or slabs, or the equivalent stones or slate, lippage, or the variations in the level of the tops of the edges of the tiles after installation, is a problem of great concern. Variations of lippage or level can be visually unacceptable, as with shadows cast on walls having strong downward, upward, or side lighting. Variation of lippage or level on floors can be dangerous as tripping hazards and as areas prone to further damage by striking raised edges.

[0004] Tile installations are tending toward larger tiles and smaller grout lines, both of which can compound lippage problems. The ANSI A108.19 standard for tile installation requires a lippage control system where any dimension of the tiles is one meter or greater. The maximum allowable lippage is $\frac{1}{32}$ inch.

[0005] Lippage control devices place compressive force on the top edges and bottom edges of adjacent tiles and force the top edges to align while accommodating variations in thickness at the bottom edges. The bottom surfaces of the tiles, including at the edges and including any lippage control device underneath the tiles, must be covered with and supported by mortar. Lippage control devices necessarily extend through the grout joint between adjacent tiles.

[0006] After the mortar has set, any parts of the lippage control device which extends up through the grout joint to the top surface of the tiles must be removed. It is undesirable to leave any scraps from the device in the grout joint, and unacceptable to leave any scraps at or near the surface of the grout. A common method for removing such parts of the lippage control device after the tiles are set is to laterally strike the parts. The striking can be done with a tool, such as a non-marking rubber mallet, but is often done with the more readily available work boot. This removal method carries a great risk of damage to the installed tile, which will frequently require cumbersome and expensive removal of damaged tile and replacement with an exactly matching tile. Many present lippage control devices either possess design features which allow the parts to become pinched or bound or blocked within the grout joint, making removal difficult and increasing the chance of damage, or features which require more cumbersome removal procedures than simply striking, sometimes requiring special tools and significant time spent on each removal.

[0007] The ANSI A108.19 standard currently recommends that any lower part of any lippage control device which remains underneath the tiles after installation should

be encapsulated with mortar. This encapsulation with mortar builds up mortar at the base of any lippage control device, and that mortar, when cured and set, can bind or block the lower parts of the device and interfere with clean removal without damage.

[0008] The smallest or thinnest grout joint allowed by the ANSI standard is $\frac{1}{16}$ " or 0.8 mm. Such a thin grout line is used in many installations. Presently known lippage control devices which have a thickness or depth in excess of $\frac{1}{16}$ " prevent installations having thinner grout lines. Although presently known lippage control devices are not necessarily designed to be used as spacers, they are often manufactured and used as spacers anyway, and are likely to become pinched or bound between the tile edges, requiring excessive force, digging, or cutting to remove, increasing likelihood of damage to the installed tiles.

[0009] U.S. Pat. No. 8,966,835 for a "Reusable Shim Shell Mechanical Edge Setting System and Method for Setting Tiles and Tuning Lippage," issued on Mar. 3, 2015 to assignee New Standards Manufacturing Co., provides for a tile aligning and lippage tuning system that uses an under tile base interstitial strap, which is connected to said under tile base, and a shim shell cap configured to slip over the connecting tab. The system also comprises of a bottom tile mating shim and a top shim.

[0010] U.S. Pat. No. 9,482,019 for a "Leveling Device for Laying Tiles or the Like," issued on Nov. 1, 2016 to assignee PROFILITEC S.P.A. SOCIO UNICO, provides for a leveling device for laying tiles, or the like, which includes a tie-rod having a support base for edge portions of one or more adjacent tiles, a projecting element from the base, which can be arranged into junctions between adjacent tiles, and a knob coupling with the projecting element and having an outlet pass-through hole for a free end of the projecting element. The knob is configured to force an edge portion of the adjacent tiles against the base to level the placement and is configured to receive and retain, by snap coupling elements, a blocking tie-rod hooking and retaining the projecting element inside a pass-through channel defined in the blocking tie-rod. The blocking tie-rod is sectioned into two coupling half-shells, embracing and constraining the projecting element when inserted in the cap, and mutually separating to leave free the projecting element when at least partially extracted from the cap.

[0011] International Publication No. 2018/052331 for a "Tile-Levelling System," published on Mar. 22, 2018 by inventor Aleksandar Jelić, discloses a tile levelling system that enables the tiling of floors, walls, and ceilings so that the resulting surface is completely flat and lippage free. The system comprises a base plate with a vertical member, which has a rectangular opening perpendicular to the base plate, and two opposite longitudinal grooves, which are V-shaped in cross section, placed below the rectangular opening on the vertical member. The upper surface of the base plate is flat. The lower surface of the base plate is corrugated with parallel ridges. Between these ridges are grooves, which ensure better adhesion of either adhesive or aggregate to the very base plate, thus making the entire construction stronger.

[0012] European Patent No. 2 549 030 for a "A Device for Correct Laying of Floor Tiles," issued on Nov. 12, 2014 to assignee Brunoplast di Eleuteri Bruno, provides for a plastic device for correct laying of floor tiles, and the device comprises of a base from which a vertical blade centrally protrudes, supporting a threaded stem in upper position and

being connected on the bottom with said base by means of a lower end with thinned section suitable to act as tearing line. The device also comprises of a knob, consisting in a cylindrical cover, starting from its upper closing wall, with a central cylindrical conduit with internal threaded walls, adapted to exactly receive, by means of a helical coupling, said threaded stem provided in upper position on said base.

[0013] U.S. Pat. No. 8,181,420 for a “Levelling Device for the Placing of Pieces for Covering Floors and Similar,” issued on May 22, 2012 to assignee Germans Boada, S.A., provides for a device that has a first lower substantially flat body, which has in its upper part a flexible post with a narrowed or weakened portion and a second upper body with the passage for receiving and securing said post. The flexible post, which has a rectangular cross section, has in its main surfaces respective mutually opposed “teeth” which are offset in the lengthwise direction and the passage of the second body has two latching pawls mutually opposed, so that in the operative position of the devices, the teeth of the flexible post engage in an alternate form with said latching pawls. The second body has a bell like hollow form with a lower widened contact mouth or border.

[0014] U.S. Pat. No. 8,079,199 for a “Tile Alignment and Leveling Device,” issued on Dec. 20, 2011 to assignee Davinci Italia/USA Group, LLC, provides for a device for aligning and leveling tiles as they are laid in floors, walls, countertops, or the like. The device has a locking assembly and a bottom plate. The components are combined with a shaft that extends from the bottom plate through the locking assembly so that the locking assembly is movable along the length of the shaft. In use, the device is placed between adjacent tiles so that the locking assembly and bottom plate hold adjacent tiles at a desired height as the setting bed dries.

[0015] U.S. Pat. No. 9,657,485 for a “Tile Leveling System,” issued on May 23, 2017 to inventor Lawrence G. Meyers, provides for a tile leveling system that includes an anchor member arranged to be positioned in a setting bed below adjacent tiles. A tensioning member extends upwardly from the anchor member and is arranged to pass between the adjacent tiles. The tensioning member is made of a metallic material and frangibly connected to the anchor member via a breakage point. A loading system is arranged to be positioned on top of the adjacent tiles and includes a drive mechanism. The drive mechanism is connected to the tensioning member and is selectively operable to secure and level the adjacent tiles between the anchor member and the loading system.

[0016] U.S. Pat. No. 9,279,259 for a “Tile Lippage Removal System,” issued on Mar. 8, 2016 to inventor William P. Russo, provides for a tile lippage removal system that includes a spacer post, a threaded cap, and an anti-friction protection plate. The spacer post includes a base member, a spacer member, and a threaded shaft. A bottom of the spacer member extends from a top of the base member. A break away connection is made between the spacer member and the base member. A bottom of the threaded shaft extends from a top of the spacer member. A plurality of grip extensions extends from an outer surface of a substantial inverted cup to allow rotation of the threaded cap. A female thread is formed in a center of the substantial inverted cup to threadably receive the threaded shaft. The anti-friction protection plate includes a round outer perimeter and a spacer opening, which is sized to receive the

spacer member. The anti-friction protection plate may be used to improve existing tile lippage removal systems.

[0017] U.S. Pat. No. 9,470,002 for a “Tile Leveller and Spacing System,” issued on Oct. 18, 2016 to assignee ME INNOVATIONS PTY LTD, provides for a tile leveller and spacer system that allows rapid and reliable spacing and levelling of tiles. The tile spacer comprises a base for locating the spacer under the tiles, and a stem that includes an alignment cross which is used to correctly space the tiles. The stem tile spacer is placed over the stem and engages with teeth on the stem. The tile leveller is pushed down over the stem to level the tiles between the base and the bottom of the tile leveller. The stem further comprises a frangible portion located between the base and the tile alignment portion, and when the stem is pulled with sufficient force, the frangible portion breaks to allow removal of the stem. A hand tool may be used to assist in this process. The hand tool can have an adjustable stop which can be set to a first setting to level the tiles, and a second setting to cause the frangible portion to break, allowing removal of the stem and leveller.

[0018] US Publication No. 2018/0355623 for a “Tile Leveling System,” published on Dec. 13, 2018 by inventor Scott Frisco, discloses a tile leveling system for leveling tiles. The system has a tile lifter whose transverse member is inserted into the grout gap, and turned to get underneath adjacent tiles. The system has a reusable clamping unit, a U-shaped, possibly flexible, wedge, which is slid through the lifter’s vertical shaft to sandwich the tiles between the transverse member and the wedge. The transverse member can be concave up, and the lifter can have two transverse members with spacers on each four-corner junction. The lifter’s handle can have a U-shaped hole, which allows a rocker tool to be used to level tiles. Another leveling tool, called a gun tool can be used with any of the lifters. Also proposed are two simpler threaded and zip-tie systems. The leveling devices can be used as needed, thus saving the cost of using them in the entire tiled area.

[0019] U.S. Pat. No. 8,429,878 for a “System and Method for Aligning and Leveling Tile,” issued on Apr. 30, 2013 to assignee New Standards Manufacturing Co., provides for a tile aligning and leveling system that uses an under tile base, a re-usable upright connecting tab, which is detachably connected to said under tile base, a re-usable flexible edge slotted cap configured to slip over the connecting tab and latch thereto as the cap is slid further down the connecting tab. The cap has an edge slot so that the connecting tab can be separated from the cap, by merely sliding the cap, so that the connecting tab passes through the slot. A flexible spring-like portion of the cap provides increased force on the tile even if undesired elongation of the connecting tab occurs.

[0020] U.S. Pat. No. 8,820,031 for a “Tile Alignment and Leveling Device,” issued on Sep. 2, 2014 to assignee Davinci Italia/USA Group, LLC, provides for a tile alignment and leveling device for aligning and leveling tiles as they are being secured to a substrate. The device includes a flexible member, a bottom plate, and an intermediate member. The intermediate member spaces the flexible member and bottom plate a predetermined vertical distance. The flexible member has a first end and a second end. The first end is pivotally combined with the intermediate member thereby allowing the flexible member to pivot between a first position and a second (downward) position. In use and in its second position, the flexible member second end exerts

force against the top of the tiles to help align and level the tiles as they are secured to the substrate.

SUMMARY OF THE INVENTION

[0021] This invention provides a tile-setting lippage and spacing control system and method for use with tiles set with mortar as floors or walls on prepared surfaces.

[0022] The tile-setting lippage and spacing control system provides spacing tabs each having a tab body with a tab opening to accommodate a wedge, a strengthened tab border, and a lower tab body meant to extend downward between tiles as a spacer. A foot unit having a footplate and footplate ridges is removably attached at the bottom of the lower tab body through tab-to-foot joins at the footplate ridges. In use, mortar is applied to a prepared surface. Tile is placed with spacing tabs placed into the grout-line gaps between tiles, with the footplate of the foot unit extending underneath both tiles and embedded in the mortar. The lower tab body extends upward between the tiles and sets the gap or grout line between the tiles. The tab body and strengthened tab border extend upward above the tops of the tiles. A wedge is pushed into the tab opening, placing downward pressure on the tops of the tiles while pulling upward against the foot unit locked underneath the tiles, creating a clamping force which holds the tops of the tiles level with each other. The tab-to-foot joins withstand the straight-line constant clamping force between the wedge and the foot unit during use, and then break when a sharp oblique force is applied, allowing removal from the gap between tiles without damage to the tiles. After the mortar cures and sets the spacing tab is removed by applying a sharp oblique force, tearing the tab body cleanly away from the foot unit which remains embedded in mortar underneath the tiles.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] Reference will now be made to the drawings, wherein like parts are designated by like numerals, and wherein:

[0024] FIG. 1 is a perspective view of three embodiments of the tile-setting lippage and spacing control system and method of the invention in use;

[0025] FIG. 2 is a perspective cutaway view of three embodiments of the tile-setting lippage and spacing control system and method of the invention in use;

[0026] FIG. 3 is an upper perspective view of three embodiments of the tile-setting lippage and spacing control system and method of the invention;

[0027] FIG. 4 is a lower perspective view of three embodiments of the tile-setting lippage and spacing control system and method of the invention;

[0028] FIG. 5 is a front view of three embodiments of the tile-setting lippage and spacing control system and method of the invention;

[0029] FIG. 6 is a side view of three embodiments of the tile-setting lippage and spacing control system and method of the invention;

[0030] FIG. 7 is a side view of three embodiments of the tile-setting lippage and spacing control system and method of the invention in use;

[0031] FIG. 8 is a front view of the spacing tab of the tile-setting lippage and spacing control system and method of the invention, in use, before removal; and

[0032] FIG. 9 is a front view of the spacing tab of the tile-setting lippage and spacing control system and method of the invention, in use, during removal.

DETAILED DESCRIPTION OF THE INVENTION

[0033] Referring to FIG. 1 and FIG. 2, the tile-setting lippage and spacing control system 10 and method for use with tiles set with mortar as floors or walls on prepared surfaces provides spacing tabs 40. The spacing tabs 40 can be used as components of lippage-control units 20 or can be used as auxiliary or standalone spacers. The lippage-control units 20 have a foot unit 60 which is permanently set into the mortar under the tiles. A spacing tab 40 is removably attached to the foot unit 60, and the spacing tab 40 extends up between two adjacent tiles, serving additionally as a spacer between the tiles, to provide for a uniform grout line. At the top level of the tiles a means of clamping, such as the wedge 30 illustrated, is used to pull the spacing tab 40 and the attached footplate upward, exerting a clamping force at the tops and bottoms of the two adjacent tiles, holding the tops of the tiles level with each other while the mortar sets. After the mortar sets the spacing tabs 40 are detached from the footplates and are removed without damaging the tiles, such as breaking the edges of the tiles, which would necessitate remedial work, as treated in detail below. The removed spacing tabs 40, from which the footplates have been removed, can optionally be reused as standalone spacers between tiles, either as auxiliary spacers alongside intact lippage-control units 20, or be used alone when no lippage-control units are required. Usually, gauged tiles or panels or slabs are used and are set with thinset mortar. Usually, a very thin grout line is used, with $\frac{1}{16}$ " or 0.8 mm being the thinnest allowed by ANSI A108.19.

[0034] Each spacing tab 40 has a lower tab body 45 having two broad faces oriented along the gap or space between the tiles, facing opposite each other, and extending downward below the top surfaces of the tiles to a foot unit 60 in use, as treated in detail below. Three embodiments of the spacing tab 40 are disclosed and illustrated, differing in the thickness profile of the lower tab body 45. A first, flat embodiment has flat faces 46 and a uniform thickness of the lower tab body 45. A second, concave embodiment has concave faces 47 and a maximum thickness at the outer edges reducing to less thickness toward the center portion of the lower tab body 45. A third, micropoint embodiment also has concave faces 47 plus micropoint structures 50 extending outward from the central portion of the lower tab body 45.

[0035] In use, the spacing tab 40 with attached foot unit 60 is placed between two adjacent tiles, on a prepared area of thinset mortar. The foot unit 60 becomes embedded in the mortar, under the tiles. The edges of the tiles make contact with the lower tab body 45 or the micropoint structures 50 on the opposing faces of the lower tab body 45, setting the spacing of the grout line between the tiles. The wedge 30 is placed through the spacing tab 40, such that the adjacent tiles are clamped between the bottom of the wedge 30 and the foot unit 60, leveling the tops of the tiles. After the mortar cures, the spacing tab 40 is removed, as treated in detail below.

[0036] Whether spacing tabs 40 are used as components of lippage-control units 20 or as standalone spacers, it is important that the tab provides the proper thickness of material to hold the tiles at the proper spacing while the

mortar sets and also provides for easy, clean, and safe removal, without risking damage to the tiles, after the mortar is set and before grouting. It is also important that the uniform spacing between the tiles be achieved at the top edges of the tiles, which can be difficult to achieve with tiles not having perfect perpendicularity and uniformity of the side faces, or where the underlying surface is not perfectly solid, smooth, and level. Although the proper thickness of material for spacing can be provided by various protrusions, such protrusions might also become stuck or blocked by set mortar making the spacer difficult to remove without damaging the set tiles. Also, where such a protrusion is located away from the top level of the tiles or is too large and extends too far away from the top edges, it becomes more likely that such a protrusion will become stuck or blocked either by mortar or by the side faces of the tiles, especially where the side faces of the tiles have some irregularity.

[0037] The spacing tabs **40** can be made of plastic material, including recycled plastic material. High-density polyethylene (HDPE) is a suitable material which has a high strength-to-density ratio and allows a controlled amount of deformation, is resistant to solvents and glues, is inherently non-marring, is somewhat self-healing, and is widely available as recycled material. The spacing tabs **40** can be manufactured by molding the plastic material, and the micropoint structures **50** can be formed in the molding process.

[0038] Referring to FIG. 3 and FIG. 4, each spacing tab **40** provides a tab body **41** which extends from the foot unit **60** upward between the tiles and above the top level of the tiles. The upper portion of the tab body **41** which extends above the top level of the tiles provides a strengthened tab border **43** which braces and strengthens the upper portion of the tab body **41** against distortion during handling, installation, curing of the mortar, and during removal. During removal, as treated below in detail, the strengthened tab border **43** also provides a suitable structure for striking the spacing tab **40** while bracing and strengthening the upper portion of the tab body **41** against unwanted deformation, crumpling, or twisting. The spacing tab **40** provides a tab opening **42** in the tab body **41** to accommodate a wedge **30**, such that in use when the wedge **30** is pressed through the tab opening **42** the two adjacent tiles are clamped between the bottom of the wedge **30** and the top of the foot unit **60**. Because the bottom of the wedge is thicker and less deformable than the foot unit **60**, the tops of the adjacent tiles will be held at the same level by the wedge **30**, and any small differences in tile thickness will be accommodated by the foot unit **60** and by the mortar which fully encases the foot unit **60** in a proper installation. As a result, after installation and curing, the tops of the adjacent tiles will be at the same level, and the bottoms of the tiles will be fully supported by cured mortar, with the foot unit **60** completely encased in the cured mortar. In preferred embodiments the material, thickness, and configuration of the tab body **41** surrounding the tab opening **42** allows for some deformation by the wedge **30** in the event that too much insertion force is applied during installation or by accident during curing, and allows for tearing or breaking under extreme strain, instead of breaking the tiles and requiring expensive remediation.

[0039] Referring additionally to FIG. 5 and FIG. 6, a foot unit **60** is removably attached to the lowest edge of the lower tab body **45**. A footplate **61** is formed from a thin piece of the same material as the tab body, with molded footplate

ridges **62** formed on the upper face of the footplate. The foot unit **60** is attached to the lowest edge of the lower tab body **45** only at the footplate ridges **62**, through tab-to-foot joints **63**. The tab body **41**, lower tab body **45**, and the footplate **61** can be formed as a single piece by molding. The tab-to-foot joints **63** are made strong enough to withstand the straight-line forces exerted by the clamping of the installed tiles between the bottom of the wedge **30** and the top of the foot unit **60**, as above. The tab-to-foot joints **63** are also made to be broken when a sharp oblique force is applied, such as when removing the spacing tab **40** after the mortar has cured, or when the spacing tab **40** is accidentally kicked or struck during installation. Removal is treated further below. The holding and the shearing properties of each tab-to-foot joint **63** is affected by the thickness of the lower tab body **45** at the lowest edge, and also by the properties of the plastic material of manufacture. The holding and shearing properties of the tab-to-foot joints **63** collectively is affected by the size and thickness of the spacing tab **40** and by the size and number of footplate ridges **62** which form the tab-to-foot joints **63**.

[0040] Referring to FIG. 7, in use of the flat embodiment having flat faces **46** on the lower tab body **45**, the space between the tiles will be set by contact along substantially the whole lower tab body **45** which is of uniform thickness in the flat embodiment. This embodiment provides strong and stable spacing which would provide advantages where tiles are likely to encounter lateral compressive strain before complete curing or in, for example, a vertical wall installation of large heavy tiles where compressive strain is unavoidable.

[0041] In use of the concave embodiment having concave faces **47** on the lower tab body **45**, but without micropoint structures **50**, the space between the tiles will be set by contact along the outer edges of the lower tab body **45**, which are the areas of maximum thickness. The tiles will make contact with two separate areas which are relatively smaller but are still more substantial than micropoint structures **50**. This arrangement provides a better smoothing out of variations in the inner surfaces of the tiles while still providing good stability against compressive strain. To the extent that the outer edges of maximum thickness begin to deform under strain, the adjacent areas of only slightly reduced thickness will act to reinforce the spacing.

[0042] In the micropoint embodiment micropoint structures **50** are provided on each of the two faces of the lower tab body **45**. Each micropoint structure **50** on one face of a spacing tab **40** has an exactly corresponding micropoint structure **50** on the opposite face of the tab. The extreme point of the micropoint structure **50** is the contact area which the tile will be placed against during installation. When two tiles are placed against the contact areas of opposite-facing corresponding micropoint structures **50** the two tiles are placed at a preset fixed distance, leaving a gap to be filled with a grout or similar material. The total spacing distance of a spacing tab **40** with micropoint structures **50** is the distance from the contact area at the extreme point of one micropoint structure **50** through the spacing tab **40** to the contact area at the extreme point of the corresponding opposite facing micropoint structure **50**. This distance will typically be at least $\frac{1}{16}$ " or 1.59 mm as required by ANSI A108.19, or $\frac{1}{8}$ " or 3.18 mm. Larger spacings are sometimes used. The size of the contact area at the extreme point of the micropoint structure **50** is very small in relation to the size

of the spacing tab 40 and the width of the gap or grout line. If the plastic material used does not have sufficient stability and resistance to compressive force, then the relatively small micropoint structures 50 might not adequately hold the intended spacing against shifting of the tiles before the mortar is fully set. HDPE material does have adequate resistance to compressive force, but does compress a small amount. An extra thickness from $\frac{1}{32}$ " to $\frac{1}{64}$ " over the desired spacing can be added during manufacture to compensate for the slight compression. An advantage of the relatively small contact areas at the extreme points of the micropoint structures 50 is that during use installing tile if any mortar were to be mistakenly applied to the contact area or to the portion of the side of the tile in contact with the contact area, the misapplied mortar will be pushed aside by the micropoint at the apex of the micropoint structure 50 and will not become trapped between the contact area and the side of the tile, which would increase the size of the gap between the tiles.

[0043] The micropoint embodiment provides the most precise control of spacing and the best adaptation to surface variations, but is the most susceptible to deform under compressive strain. If the micropoint structures 50 collapse the tiles will then be supported at the outer edges of maximum thickness of the lower tab body 45.

[0044] Referring to FIG. 8 and FIG. 9, in use, after the mortar is cured, the foot unit 60 is permanently embedded in the mortar underneath the tiles. The spacing tab 40 is then separated from the foot unit 60 and is removed from the space between the tiles by the application of a sharp oblique force such as hitting with a rubber mallet. The strengthened tab border 43 around the upper portion of the tab body 41 prevents the force from crumpling or collapsing the tab body, and also accommodates poorly aimed blows by directing the force toward the proper direction. The sharp oblique force shears and tears the spacing tab 40 from the foot unit 60 at the tab-to-foot joins 63. The spacing tab 40 is easily and safely removed from the space between the tiles without damage to the tiles. If any cured mortar is built up around the lower tab body 45, the spacing tab 40 will pivot around the mortar and out of the grout line. An accidental application of force, such as kicking or striking the installed spacing tab 40, will also cause the spacing tab 40 to shear away from the foot unit 60, without damaging the tiles.

[0045] Many other changes and modifications can be made in the system and method of the present invention without departing from the spirit thereof. We therefore pray that our rights to the present invention be limited only by the scope of the appended claims.

We claim:

1. A tile-setting lippage and spacing control system for use with tiles set with mortar as floors or walls on prepared surfaces, the tile-setting lippage and spacing control system comprising:

- (i) a spacing tab adapted to set a spacing between tiles, having two opposing faces oriented along the gap between tiles, and having a tab body having an upper and lower orientation with a lower tab body extending downward between the tiles, and with a strengthened tab border surrounding the upper portion of said tab body, and having a tab opening in said tab body;
- (ii) a foot unit having a footplate having footplate ridges, removably attached to the lowest edge of said lower tab

body through tab-to-foot joins at each said footplate ridge; and

- (iii) a wedge adapted to fit said tab opening;

where, in use, said spacing tab with attached said foot unit is placed between tiles being set into uncured mortar, setting the size of the gap between tiles, and with both tiles being clamped between said wedge placed through said tab opening at the top of the tiles and the foot unit at the bottom; and

where, in use, said spacing tab can be separated from said foot unit at said tab-to-foot joins by application of a sharp oblique force to said strengthened tab border, and can be removed from the gap between tiles without damage to the tiles.

2. The tile-setting lippage and spacing control system of claim 1, where said spacing tab is further made of a plastic material.

3. The tile-setting lippage and spacing control system of claim 1, where said spacing tab is further made of a recyclable plastic material.

4. The tile-setting lippage and spacing control system of claim 1, where said spacing tab is further made of high-density polyethylene.

5. The tile-setting lippage and spacing control system of claim 1, where said spacing tab is further adapted for re-use as a spacer in later tile installations.

6. The tile-setting lippage and spacing control system of claim 1, where said lower tab body further comprises flat faces.

7. The tile-setting lippage and spacing control system of claim 1, where said lower tab body further comprises concave faces.

8. The tile-setting lippage and spacing control system of claim 1, where said lower tab body further comprises micropoint structures.

9. A tile-setting lippage and spacing control system for use with tiles set with mortar as floors or walls on prepared surfaces, the tile-setting lippage and spacing control system comprising a lippage-control unit adapted to hold the top edges of adjacent tiles level with each other and to set the spacing between tiles during installation and setting of the mortar, comprising:

- (i) a spacing tab adapted to set a spacing between tiles, having two opposing faces oriented along the gap between tiles, and having a tab body having an upper and lower orientation with a lower tab body extending downward between the tiles, and with a strengthened tab border surrounding the upper portion of said tab body, and having a tab opening in said tab body;
- (ii) a foot unit having a footplate having footplate ridges, removably attached to the lowest edge of said lower tab body through tab-to-foot joins at each said footplate ridge; and
- (iii) a wedge adapted to fit said tab opening;

where, in use, said spacing tab with attached said foot unit is placed between tiles being set into uncured mortar, with said micropoint structures setting the size of the gap between tiles, and with both tiles being clamped between said wedge placed through said tab opening at the top of the tiles and the foot unit at the bottom; and where, in use, said spacing tab can be separated from said foot unit at said tab-to-foot joins by application of a

sharp oblique force to said strengthened tab border, and can be removed from the gap between tiles without damage to the tiles.

10. A tile-setting lippage and spacing control method comprising:

- (i) providing a tile-setting lippage and spacing control system comprising:
 - (a) a spacing tab adapted to set a spacing between tiles, having two opposing faces oriented along the gap between tiles, and having a tab body having an upper and lower orientation with a lower tab body extending downward between the tiles, and with a strengthened tab border surrounding the upper portion of said tab body, and having a tab opening in said tab body;
 - (b) a foot unit having a footplate having footplate ridges, removably attached to the lowest edge of said lower tab body through tab-to-foot joins at each said footplate ridge; and
 - (c) a wedge adapted to fit said tab opening;
- (ii) applying mortar to the prepared surface;
- (iii) placing said spacing tab between two adjacent tiles, setting the size of the gap between tiles;
- (iv) placing said wedge through said tab opening, with the tiles being clamped between said wedge and said foot unit;
- (v) allowing curing and setting of the mortar,
- (vi) removing said spacing tab from the grout joint by application of a sharp oblique force; and
- (vii) grouting the grout joint.

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